Maintain Airway, Breathing and Circulation

**Administer Oxygen**
10-15L/min via NRB Mask
**Regardless of SpO2**

EMR

**Position Patient**
In supine or Left Lateral Position

Consider ALS Back-Up

**Administer Crystalloid Fluid**
**Adult:** 500ml IV
**Pediatric:** 10ml/kg IV

EMT

Treat associated symptoms as per appropriate CPG

Transport dive computer and diving equipment with patient, if possible

Continuous Monitoring of Vitals Signs

Coordinate to facility with hyperbaric chamber

**Suspected Diving Emergency Symptoms**

- Recent diving history with symptoms such as:
  - Shortness of breath, chest pain, coughing
  - Dizziness, confusion, numbness, weakness
  - Joint or muscle pain
  - Visual changes, tinnitus, fatigue, collapse
  - Symptoms developing within 24 hours of a dive

## Patient Care Goals

1. Recognize diving-related injuries early and prevent deterioration
2. Support airway, breathing, and circulation
3. Provide high-concentration oxygen to reduce bubble size and hypoxia
4. Maintain supine positioning unless contraindicated
5. Identify serious complications that require hyperbaric therapy
6. Provide safe, urgent transport with early hospital notification
7. Minimize further physical exertion and time delays

## Patient Inclusion & Exclusion Criteria

**Inclusion Criteria**

- Recent diving history with symptoms such as:
  - Shortness of breath, chest pain, coughing
  - Dizziness, confusion, numbness, weakness
  - Joint or muscle pain
  - Visual changes, tinnitus, fatigue, collapse
- Symptoms developing within 24 hours of a dive

**Exclusion Criteria**

- Patients without a history of diving (SCUBA) within the last 48 hours.

## Key Patient Assessment Points

1. **Time of Last Dive & Dive History**
   - Time and number of dives
   - Depth and duration of dive
   - Use of compressed gas (e.g., scuba/nitrogen)
   - Rapid ascent or missed safety stops
   - Use of flying or altitude after diving
2. **Airway**
   - Assess for patency, gurgling, or vomiting
3. **Breathing**
   - Assess for dyspnea, cough, chest tightness, or tachypnea
   - Auscultate for wheezing, crackles, or decreased breath sounds
   - Monitor $SpO_2$ — aim for ≥94%
4. **Circulation**
   - Check HR, BP, perfusion, skin color
   - Look for arrhythmias, bradycardia, or hypotension
   - Cyanosis may indicate serious gas embolism
5. **Disability (Neurological)**
   - Use GCS or AVPU to assess mental status
   - Assess for:
     - Confusion, headache, dizziness, slurred speech
     - Paralysis, numbness, seizures, visual or hearing disturbances
     - Ataxia or difficulty walking
6. **Exposure**
   - Evaluate for rash or skin mottling (cutis marmorata)
   - Joint or back pain suggests decompression sickness (DCS)
   - Check for signs of barotrauma (e.g., facial bruising, ear bleeding)
7. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority
       - Neurological symptoms (paralysis, confusion, seizure)

- Chest pain, dyspnea, cyanosis
- $SpO_2$ <90% or GCS <13
- Suspected arterial gas embolism (AGE)
- Collapse or unconsciousness after surfacing
- Yellow Priority
  - Joint/muscle pain or mild dizziness
  - Alert and oriented with stable vitals
  - Symptoms improve with oxygen but not fully resolved

## Patient Management & Interventions

1. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG
   - Activate early for all Red Priority or neurological/respiratory cases
2. **Oxygen Therapy**
   - Administer 100% oxygen via NRB mask regardless of $SpO_2$ if a SCUBA incident is suspected.
   - High-concentration oxygen helps reduce nitrogen/gas bubbles
   - Do not withhold oxygen if patient appears stable
   - Continue oxygen throughout transport
3. **Positioning**
   - Keep patient supine, unless vomiting or in respiratory distress
   - Do not elevate legs or place head-down; may worsen embolism or cerebral edema.
   - Lateral (recovery) position if unconscious or vomiting
4. **Fluid Administration**
   - If hypotensive or showing signs of dehydration, administer IV Fluids as per Universal Care CPG.
   - Avoid excessive fluids unless guided by ALS/TeleEMS
5. **Pain Management**
   - Provide pain relief as per Pain Management CPG.
6. **Avoid Worsening Triggers**
   - Avoid exertion, standing, or walking unassisted
   - Keep patient warm and calm
   - Helicopter evacuation unless unavoidable
7. **Transport**
   - Transport all symptomatic patients urgently
   - Notify receiving hospital early; suspect DCS or AGE
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Specific Patient Reassessment Points:
     - Level of consciousness (GCS or AVPU):
       - Track for any improvement or deterioration — repeated confusion, drowsiness, or decreased responsiveness may indicate worsening embolism or cerebral decompression sickness.
     - Neurological examination:
       - Reassess limb strength, movement, and sensation every 5–10 minutes.
       - Check speech clarity, facial droop, and coordination.
       - Document any new or worsening deficits.
     - Pain assessment:
       - Ask about joint, back, or chest pain and document changes. Sudden or migrating pain may indicate evolving decompression sickness.
     - Response to oxygen therapy:
       - Note improvement in symptoms (e.g., clearer speech, reduced dizziness, better $SpO_2$)
       - Lack of improvement may suggest severe gas embolism or delayed response to therapy; continue high-flow oxygen and expedite transport.

## Patient Safety Considerations

- Always suspect DCI or AGE in divers with any neuro, resp, or unexplained symptoms after ascent.
- 100% $O_2$ is the single most important intervention prehospital. Early oxygen can prevent worsening of decompression illness
- Do not allow walking or physical exertion; increases gas migration
- Avoid placing in Trendelenburg (head-down)
- Prepare for delayed symptom onset; symptoms can appear hours later
- All suspected AGE or serious DCS should go to a hospital with hyperbaric capability.
    - Coordinate with Dispatch 2 to confirm that the receiving facility has a hyperbaric chamber.
    - Note: Consider Al Zahra hospital.
    - If a hyperbaric chamber is not immediately available, the patient may need to be management at an appropriate emergency department first, while waiting for chamber to be available.

## Clinical Insights & Additional Guidance

- Decompression sickness (DCS) is caused by nitrogen bubbles forming in tissues after a dive.
- Arterial gas embolism (AGE) occurs when air enters the bloodstream, often due to lung over-expansion injury.
- Symptoms can appear minutes to hours after surfacing.
- Never delay transport, definitive treatment is hyperbaric oxygen therapy (HBOT).
- Mild symptoms may worsen if untreated.
- Children and first-time divers are at risk due to poor dive control.
- Always consider spinal involvement in unconscious divers.

## Documentation

- Identification: Age, gender, weight
- Medical complaint: Diving-related symptoms
- Information: Time of dive, depth, symptoms, previous episodes
- Signs: Vitals, $SpO_2$, GCS, neuro findings
- Treatment: Oxygen, positioning, ALS request, fluids if given
- Allergies: Relevant medication or environmental allergies
- Medications: Any meds taken, or given prehospital
- Background: Prior dive experience, medical history
- Other: Dive profile:
    - Time of last dive
    - Dive profile
    - Onset and type of symptoms
    - Response to oxygen

## Key Performance Indicators

1. Administer oxygen within 5 minutes of arrival at patient.
2. Document dive time and depth in ePCR.

## References


- Vann, R.D., Butler, F.K., Mitchell, S.J., and Moon, R.E. (2011). Decompression illness. Lancet, 377(9760), pp.153-164. https://doi.org/10.1016/S0140-6736(10)61085-9
- American Heart Association. (2020). PALS Provider Manual. Dallas: AHA.
- Greaves, I. and Porter, K. (2007). Oxford Handbook of Pre-Hospital Care. Oxford: Oxford University Press.
- Divers Alert Network. (2024). Emergency Assistance Guidelines. Available at: https://www.diversalertnetwork.org
- DCAS. (2024). Clinical Practice Guidelines – Version 02.